

FINAL SEMINAR · OCTOBER 2026

Multi-Page Visually Rich Document Understanding

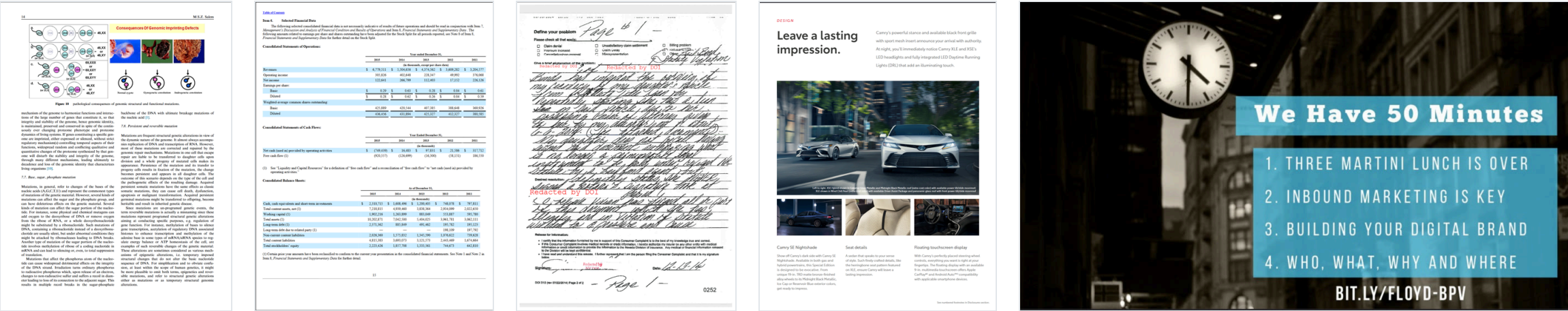
A survey of system design and an empirical attribution of failure

Lewei Xu

Supervised by Dr. Yihao Ding · Dr. Daochang Liu

Visually Rich Documents

A visually rich document is one whose meaning is carried jointly by text, layout, and visual content rather than by text alone.



Representative pages from MMLongBench-Doc

Text is small and dense

Typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.

Layout contains semantic information

A number means one thing in a table's first column and another in its second, so position must be interpreted rather than read.

Reading order must be recovered

A page is a two-dimensional arrangement rather than a linear stream, so the order cannot be assumed.

General-Domain MLLMs and VRDU

Frontier general-domain MLLMs read a rendered page directly from pixels and handle single-page and short multi-page inputs reliably when resolution is sufficient. Even so, the products built on them rarely rely on page images alone.

DIRECTLY IN CONTEXT

"The full extracted text, plus page images for PDFs within the visual limits, is loaded into my context. I see all of it, with no retrieval step."

ON DISK, VIA FILE TOOLS

"The content isn't in my context yet. I read it on demand, either the whole thing [...] or selectively [...]. That's my choice per task."

RETRIEVAL

"This only applies to Projects, where knowledge that exceeds the context window is indexed and I search it rather than ingest it."

Claude, asked how it handles an uploaded PDF.
Lightly elided; emphasis in the labels is mine.

Already a document pipeline

In their web interfaces and coding agents, both Anthropic and OpenAI parse an uploaded PDF for text and render it to per-page images, placing both in the model's context. That is a document-specialised pipeline, albeit a very simple one.

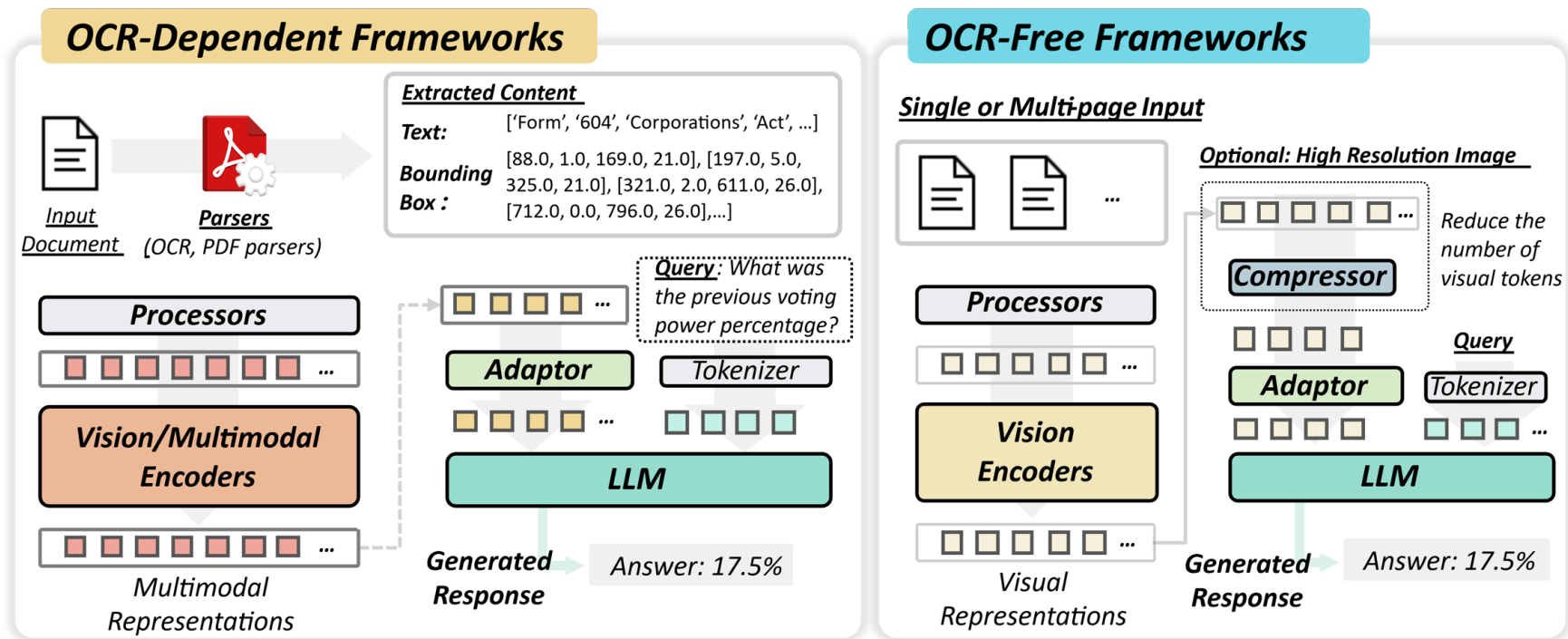
Its limits arrive quickly

Uploading a large document or several smaller ones soon fills the context window, and more complex document types often fail to be read at all.

The frontier is how general-domain MLLMs can be leveraged, together with the machinery built around them, to answer questions over documents the backbone **cannot process directly**. That machinery is the subject of MP-VRDU.

Single-Page VRDU Overview

The single-page setting is largely solved, and multi-page systems inherit their per-page machinery from it.



OCR-dependent and OCR-free architectures for MLLM-based VRDU. Adapted from Ding et al.

OCR-dependent

Parse the page first, recovering text with its bounding-box coordinates. Captures structure without heavy visual pretraining, but ties accuracy to the parser's.

OCR-free

A vision encoder reads the rendered page from pixels. Handles handwriting and irregular layouts, but needs high resolution, so compression usually follows.

Challenges of the Multi-Page Setting

- **The document exceeds the context window.** Long multi-page documents routinely outrun what a backbone can hold, so the system must select which parts of the document reach the MLLM at all.
- **Scaling the context does not resolve it.** Long-context models attend poorly to evidence buried mid-input, and memory and compute grow with sequence length, so the achievable context is bounded by hardware.
- **Every required evidence unit must be located.** Answering often means integrating evidence from several parts of the document at once, and missing any one of them is enough to answer incorrectly or not at all.
- **Cross-page structure is not recoverable in isolation.** Section hierarchy and cross-references cannot be read off any single part of the document, yet questions often need them to gather the evidence in the first place.

Seminar Overview

Two papers came out of this Honours project. The rest of the seminar follows them in turn.

1 **Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding**

Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

Main Conference of EMNLP 2026

NEXT MP-VRDU organised through evidence management: a taxonomy of four architectural families, multimodal representation, and inference-time and training strategies.

2 **Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution**

Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

Submitted to EACL, under review

THEN Failure attributed to three loci, representation, selection, and reasoning, through controlled interventions on a single pipeline.